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To cite this article: Kevin J. Blair , Alexa Monroy , Jordan M. Rook , Ethan Wood , J. Esteban Foianini , Adil H. Haider , Mamta Swaroop & Marissa A. Boeck (2020) What is trauma? Qualitatively assessing stakeholder perceptions in Santa Cruz de la Sierra, Bolivia, Global Public Health, 15:9, 1364-1379, DOI: [10.1080/17441692.2020.1761424](https://doi.org/10.1080/17441692.2020.1761424)

To link to this article: <https://doi.org/10.1080/17441692.2020.1761424>

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What is trauma? Qualitatively assessing stakeholder perceptions in Santa Cruz de la Sierra, Bolivia

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ABSTRACT

Addressing the burden of injury in low-resource settings requires development of trauma systems. This study aimed to describe perceptions of trauma in Santa Cruz, Bolivia to better inform strategies for trauma system development. In 2015–2016, we conducted 16 individual and 11 group interviews with key stakeholders involved with or exposed to trauma. A pile sorting activity showed participants pictures of injury mechanisms to explore perceptions of trauma. Responses were analysed for themes using content and discourse analysis. Among 27 interviews, six were with physicians, seven with first responders, three with community members, and 11 with trauma patients. Pictures commonly categorised as trauma depicted a road traffic incident (92.6%), fall (88.9%), gunshot wound (88.9%), and burn (85.2%). Fewer respondents stated intoxication (51.9%) or drowning (40.7%) were trauma. Coding of responses revealed five themes: trauma definition, mechanism, physical injury, management, and psychological trauma. Medical personnel focused more on trauma as mechanism, physical injury, and management, whereas laypersons commonly described trauma as psychological. Varied understanding of what represents trauma could influence trauma registry data collection. Laypersons' focus on psychological trauma may affect use of designated trauma care hospitals. These viewpoints must be considered when designing policies and interventions for trauma system strengthening.

ARTICLE HISTORY

Received 22 January 2020
Accepted 13 April 2020

KEYWORDS

trauma; qualitative; Bolivia; language; meaning

Introduction

A growing body of research over the past decade has focused on the global burden of injury and the need to develop trauma systems. Nearly six million annual deaths are attributable to injury, with 90% occurring in low- and middle-income countries (LMICs) (World Health Organization [WHO], 2008, 2010a). This number is expected to rise, partly because of a projected increase in road traffic incidents (RTI) (Mathers & Loncar, 2006). Interventions to address the burden of injury in LMICs have focused on the development and strengthening of pre-hospital and hospital trauma systems (WHO, 2010b). The World Health Assembly recently passed resolution 72.31, which calls on

the WHO member states to consider emergency and trauma care services as part of universal health coverage (WHO, 2019b).

Development of trauma systems often begins with the implementation of trauma registries in order to gather data on the burden of injury in a region, as well as to provide a means of measuring intervention effects (Bommakanti et al., 2018; Juillard et al., 2009; Moore & Clark, 2008). Success of these initial stages of trauma system development is dependent on a shared definition and understanding of injury and trauma, which could be impacted by differing cultural or linguistic contexts (Gjersing et al., 2010). It is critical for stakeholders involved in trauma system development to have a clear comprehension of injury and trauma, but there is a lack of literature examining the understanding of these topics across LMICs.

Injury occurs when an external agent, through the transfer of energy, causes physical damage to the body (American College of Surgeons Committee on Trauma [ACS-COT], n.d.). The WHO describes various injury mechanisms, including RTIs, suicide, homicide, falls, drowning, poisoning, fires, and war (Global Burden of Disease [GBD] Study Online Tools, 2017; WHO, 2010a). While the definition of injury is well accepted, there is a lack of global consensus on which inclusion criteria should be utilised when implementing trauma data registries in LMICs (ACS-COT, n.d.; Bommakanti et al., 2018; Haider et al., 2014; O'Reilly et al., 2013). The National Trauma Data Bank (NTDB), a nationwide collection of trauma registry data in the United States, uses inclusion criteria based on the International Classification of Diseases (ICD) injury diagnostic codes rather than mechanism, and notably excludes superficial injuries as well as those related to drowning and poisoning (ACS-COT, 2018). Some LMIC trauma registries comply with the NTDB inclusion criteria, including the registry of the Panamerican Trauma Society (Panamerican Trauma Society [PTS], n.d.). Others have chosen inclusion criteria which more broadly capture all injuries, including drowning and poisoning, in an effort to highlight the complete burden of injury in a region (Boeck et al., 2016; Nwanna-Nzewunwa et al., 2018; Ringdal et al., 2008; Schultz et al., 2007).

The Plurinational State of Bolivia (Bolivia) is a lower-middle income country with a population of just over 11 million located in central South America (Figure 1) (C.I.A. World Fact Book, 2015; WHO, 2019a). Although it is suspected that Bolivia suffers from a high burden of injury, specifically from RTIs, Bolivia lacks an integrated trauma system and data registry (WHO, 2015). Recently, efforts to develop a more cohesive trauma system, including a pilot trauma registry, have taken place in Santa Cruz de la Sierra (Santa Cruz), the country's most populous city (Boeck et al., 2016, 2018; Markin et al., 2013). This study aimed to determine stakeholder perceptions and understanding of trauma in Santa Cruz, Bolivia in order to better inform trauma registry implementation, as well as future trauma system development and injury prevention efforts in Bolivia and other LMICs.

Materials and methods

Setting and participant selection

A series of individual and group interviews were conducted in Santa Cruz with key healthcare system and community stakeholders who were involved with or exposed to trauma. Planning and implementation of this study involved close coordination and guidance from local leaders within the government and healthcare system in Santa Cruz. One of the authors (E.F.) and other local leaders identified participants for inclusion. Participants were purposively sampled from four key stakeholder groups: physicians, first responders, community members, and trauma patients. Physicians were from two public, tertiary care hospitals in Santa Cruz involved in the trauma registry project: Hospital San Juan de Dios (HSJD) and Hospital Japonés (HJ). First responders were from both governmental and non-governmental organisations, as well as a volunteer group (Table 1). Community members were either affiliated with a self-defence training group, or were neighbourhood political activists. Lastly, trauma patients were selected from four hospitals involved in the trauma registry



Figure 1. Map of Santa Cruz, Bolivia.

project: HSJD, HJ, Clínica Foianini (CF, a private hospital), and Hospital Plan 3000 (HPTM, a second level public hospital).

Physicians, first responders, and community members were contacted prior to the interview day to ensure availability and willingness to participate. The decision to conduct interviews individually or as a group, and the number of sessions held, was influenced by participant interest, and researcher and participant availability. Trauma patients were selected by convenience sampling and approached during their stay in the respective hospital emergency department. Verbal consent was obtained prior to interviews.

Study design

Semi-structured interview guides were developed prior to the initiation of the study (Colorafi & Evans, 2016). Guides were developed in English and subsequently translated into Spanish. Each interview included a pile sorting activity, during which participants were shown pictures depicting six WHO injury categories, and were asked whether or not each represented trauma. [Figure 2](#)

Table 1. First responder groups of Santa Cruz, Bolivia.

Name	Type	Funding	Types of emergencies, per mission statements
Departamento de Emergencias Municipales (DEM)	Public	City	Pre-hospital care/transport, fires, search and rescue.
Fundaso!	Volunteer	Fundraising	Fires, pre-hospital care/transport, incidents involving dangerous substances.
Medicina Integral Activa (MIA)	Private	Private	Subscriber-based, pre-hospital care and transport for medical emergencies, specialising in intensive care. Offers other medical services.
Sistema Integrado de Emergencias Médicas (SISME)	Public	City	Pre-hospital care and transport.
Bomberos de la Policía	Public	Federal	Specialization in fires, though also respond to RTIs and other emergencies.
Unidad de Emergencias Médicas (UDEM)	Private	Private	Subscriber-based, pre-hospital care and transport, fires. Offers services at a large corporation level.

presents the images showing the mechanisms of fall, RTI, burn, drowning, gunshot wound (GSW), and intoxication. A change was made to the GSW picture during the course of the study at the recommendation of Bolivian partners. Interviews also included questions about stakeholder impressions of the current emergency response and trauma system in Santa Cruz; those responses are not included in this study.

Interviews took place in November and December of 2015 (physicians, first responders, and community members) as well as June and July of 2016 (trauma patients), and were led by K.J.B., A.M., or J.M.R. in Spanish. K.J.B. and J.M.R. are male American citizens fluent in Spanish as a second language; A.M. is a female American and Guatemalan citizen fluent in Spanish as her first language. All three interviewers were medical students visiting Bolivia for a short period of time to conduct these interviews, as well as to participate in other trauma-related projects. Study completion was informed by resource constraints, though thematic saturation was also achieved. For individual interviews, the pile sorting activity took a mean of 5 min (min) 1 s (sec), with a mean total interview time of 30 min 50 sec. Group interviews had a mean activity time of 12 min 12 sec with a mean total time of 63 min 22 sec.

Refreshments were provided to all participants except those who were trauma patients. Community members received travel compensation, while other groups were interviewed at their place of work or during hospital admission, negating transportation costs. All interviews were conducted anonymously besides the general identification of stakeholder type.

IRB approval was obtained from Northwestern University, in addition to approval from the directors of each participating hospital or institution. The study is reported in accordance with the Standards for Reporting Qualitative Research (O'Brien et al., 2014).

Data analysis

Audio-recorded interviews were transcribed verbatim in Spanish, and translated into English by K.J.B., A.M., J.M.R., and E.W., all of whom are fluent in Spanish and received training in the qualitative interviewing process. Transcripts were analysed according to content analysis (Leech & Onwuegbuzie, 2008) and discourse analysis (Starks & Trinidad, 2007) using Dedoose software (Dedoose Version 8.2.14, 2019), a qualitative research platform. Interviews were systematically coded to identify key themes. Six of the authors, K.J.B., A.M., J.M.R., E.W., M.S., and M.A.B. were involved in the coding of the first transcript to establish a codebook. Demographics for K.J.B., A.M., and J.M.R. are described above. E.W. is a male American fluent in Spanish as a second language, who was an undergraduate student at the time of the study. M.S. is a female British, American, and overseas Indian citizen and trauma surgeon conversant in Spanish as a third language. M.A.B. is a female American citizen fluent in Spanish as a second language who, at the time of

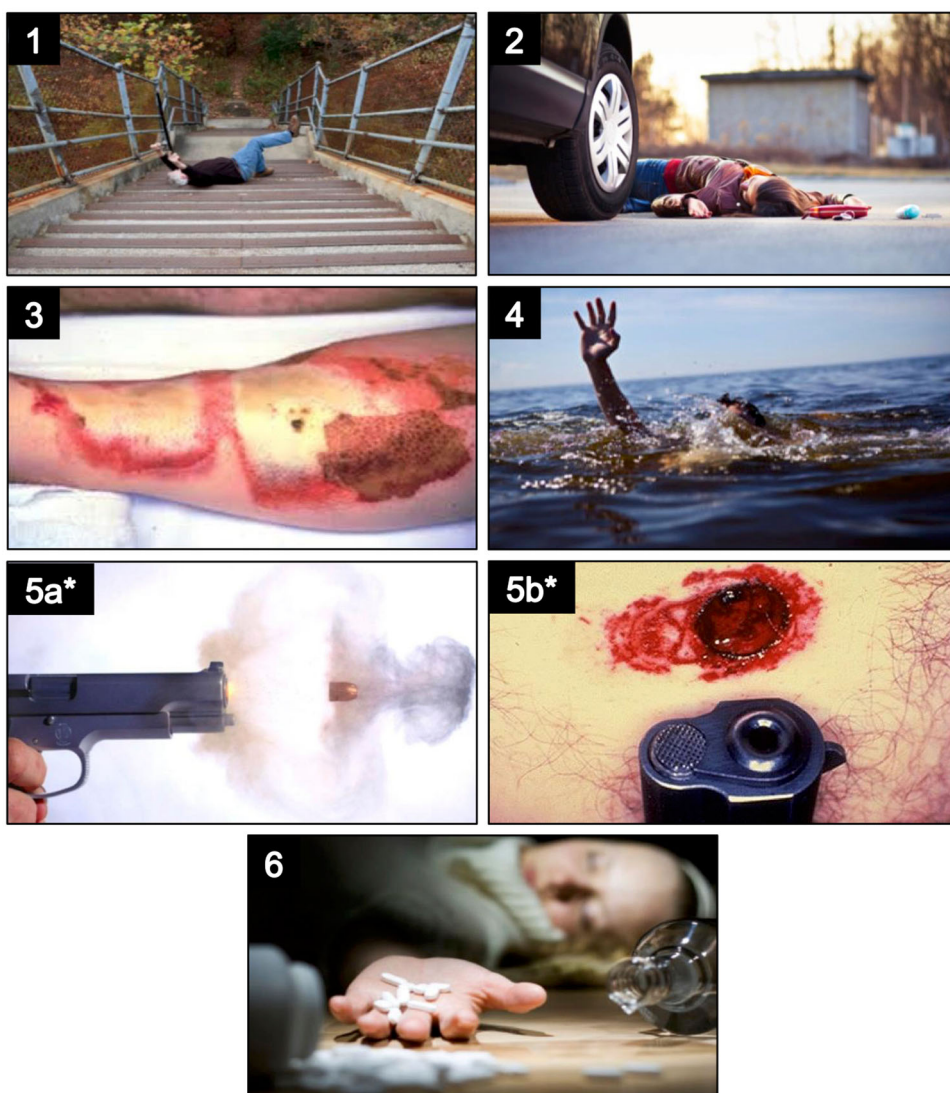


Figure 2. Pictures used in pile sorting activity. These six pictures were chosen to represent the following WHO injury categories: (1) fall, (2) road traffic incident (RTI), (3) burn, (4) drowning, (5) gunshot wound (GSW), (6) intoxication/poisoning. *Image 5a was changed to 5b partway through the study for enhanced clarity.

this study, was a surgery research fellow living in Bolivia full-time as the project lead. All coders achieved an individual kappa score of 0.75 or greater on an inter-rater reliability test to ensure synchrony in code interpretation and application. Remaining transcripts were coded by K.J.B., A.M., J.M.R., and E.W.

Common themes were first identified by frequency using content analysis. Subsequently, transcripts were re-analysed using discourse analysis to understand how common themes connected with participant demographics and narratives as well as the healthcare system. This approach was chosen due to this study's focus on stakeholder understanding of trauma as a word and concept within the medical field, and the goal of informing future policy and interventions to reduce injury related mortality. Our analysis was planned and implemented under the guidance of two experienced qualitative researchers, as mentioned in acknowledgements.

Results

Participant demographics

Twenty-seven interviews were conducted during the study period: six with physicians, seven with first responders, three with community members, and 11 with trauma patients. Sixteen of these interviews were conducted individually, and 11 as group interviews. Two group interviews were conducted with physicians, seven with first responders, and two with community members. A total of 62 individuals were interviewed, with nearly three-quarters being male (Table 2). Physicians were either attendings or resident/house officers completing specialty training in medicine or surgery at HSJD or HJ. Twenty-six individuals were employed first responders from one of the governmental or non-governmental organisations, and seven were volunteer first responders with full-time jobs in addition to their volunteer responsibilities. For analysis purposes the nine community members and 11 trauma patients were grouped together as laypersons.

Picture pile sorting activity

The picture depicting an RTI was the image most frequently described as trauma, with participants in 25 (92.6%) interviews responding affirmatively. This was followed by the fall ($n = 24$, 88.9%), GSW ($n = 24$, 88.9%), and burn ($n = 23$, 85.2%) pictures. Responses were divided for the pictures depicting intoxication and drowning, with only 14 (51.9%) and 11 (40.7%) interviews, respectively, saying the images represented trauma. The participants who considered the intoxication and drowning pictures as trauma were predominately laypersons; no physician considered intoxication or drowning as trauma. Table 3 presents the pile sorting categorizations for each picture. There were a few instances in which a response was not collected for a given image, often because of interruptions. Additionally, there were two instances during group interviews in which group members had differing responses.

Table 2. Demographic characteristics of interview participants.

Demographics	Participants <i>N</i> (%) (<i>N</i> = 62)
Gender:	
Male	45 (72.6)
Female	17 (27.4)
Interview type:	
Individual	16 (25.8)
Group*	46 (74.2)
Stakeholder group:	
Physician	9 (14.5)
Male	8 (12.9)
Female	1 (1.6)
First Responder, public	20 (32.3)
Male	15 (24.2)
Female	5 (8.1)
First Responder, private	6 (9.7)
Male	5 (8.1)
Female	1 (1.6)
First Responder, volunteer	7 (11.3)
Male	5 (8.1)
Female	2 (3.2)
Community Member [†]	9 (14.5)
Male	3 (4.8)
Female	6 (9.7)
Trauma Patient [†]	11 (17.7)
Male	9 (14.5)
Female	2 (3.2)

*Two physician, seven first responder, two community member group interviews.

[†]Community members and trauma patients are grouped together as laypersons for analyses.

Table 3. Pile sorting categorisation of pictures as trauma.

Picture number*	Mechanism depicted	Yes N (%)	No N (%)	Mixed N (%)	Not answered N (%)
1	Fall	24 (88.9)	0 (0.0)	0 (0.0)	3 (11.1)
2	RTI	25 (92.6)	2 (7.4)	0 (0.0)	0 (0.0)
3	Burn	23 (85.2)	3 (11.1)	1 [‡] (3.7)	0 (0.0)
4	Drowning	11 (40.7)	14 [†] (51.9)	0 (0.0)	2 (7.4)
5	GSW	24 (88.9)	2 (7.4)	0 (0.0)	1 (3.7)
6	Intoxication	14 (51.9)	12 [†] (44.4)	1 [‡] (3.7)	0 (0.0)

*See Figure 2 for corresponding pictures.

[†]Six respondents for both drowning and intoxication stated the picture did not depict trauma unless there was an injury as a result of a concurrent mechanism.

[‡]A group with differing responses among members, some responding 'No' and others 'Yes'.

Abbreviations: Road Traffic Incident (RTI), Gunshot Wound (GSW).

In total, we applied 49 unique codes 891 times across the 27 interviews. Codes were organised into five major themes (Table 4): trauma definition, mechanism, physical injury, management, and psychological trauma. The 25 codes with at least 10 applications are presented as sub-themes in Table 5, all of which were mentioned in at least 10% of interviews. A complete list of codes is available as supplemental material.

Trauma definition

Nearly all respondents made specific statements regarding their definition or understanding of trauma. Some participants defined trauma by severity; they affirmed injuries as trauma 'because the injuries are serious,' or denied injuries as trauma because the 'burn ... only [affects] the skin'. Certain pictures were not categorised as trauma because the situation was considered an accident by participants. One layperson explained the fall 'wouldn't really be a situation of trauma, because ... [it is] like an accident that can occur to anyone'. Likewise, a surgeon stated the drowning image 'doesn't fit much in trauma, it's an accident'. A small number referenced the concept of trauma being due to 'an external agent,' which 'provokes harm to the person' and 'has affected you in some way'. Others used the temporality of an injury or its associated recovery to define trauma, with statements such as '[it's a] trauma because it's an acute occurrence,' or 'psychological trauma takes longer to get rid of ... she won't forget anymore what happened'.

A majority of respondents tried to define trauma when discussing each of the six pile sorting pictures. Nearly all of those who provided a definition of trauma when discussing the fall, RTI, burn, and GSW pictures thought each of these represented a trauma. For the drowning and intoxication pictures, however, half of those who thought the image did not represent trauma reasoned they did not depict a physical injury.

Mechanism

All 27 interviews discussed mechanism. Many referenced the specific mechanisms depicted in each picture, with statements such as 'all burns are injuries ... due to physical means that are considered trauma' or 'injury by firearm generally is a trauma'. A physician explained, 'Traffic accidents don't merit much discussion' because you can have 'trauma on the level of the thorax, cervical trauma, or intra-abdominal trauma,' suggesting RTIs always qualify as traumas because of the various physical injuries which can result.

Participants frequently used a picture's depicted injury mechanism to explain their understanding of trauma. For example, respondents referenced the staircase in the fall picture as a 'frequent place ... of trauma,' especially when 'it's an elderly person'. Similarly, the staircase frequently elicited a reference to height of the fall, with participants clarifying 'due to the procedures that drive falls

Table 4. Themes from picture pile sorting responses.

Theme	Interviews with theme N (%) N = 27	Total theme applications*	Physician applications N (%)	Representative quote from physician	First responder applications N (%)	Representative quote from first responder	Layperson applications N (%)	Representative quote from layperson
Trauma Definition	26 (96.3)	166	22 (13.2)	'Patient is drowning ... but it's not a trauma. Unless the patient has fallen, has fallen from a height, approximately'.	74 (44.6)	'... I believe that yes you could say that it is a trauma because in the end it is an external agent that ... provokes harm to the person'.	70 (42.2)	'... for me it wouldn't really be a situation of trauma, because the man is falling down the stairs, like an accident that can occur to anyone'.
Mechanism	27 (100.0)	355	64 (18.0)	'... all burns are injuries. It's an, an injury due to ... physical means that are considered trauma, no?'	154 (43.4)	'The kinematics of the fall ... The height, due to the procedures that drive falls from a height, where we catalogue them as, as a possible situation of trauma'.	137 (38.6)	'I'm not sure if burns can be treated like a trauma, but possibly. I think that yes'.
Physical Injury	24 (88.9)	156	25 (16.0)	'Obviously, it's a trauma, no? Because she could have multiple traumas, and injuries in this part of her body'.	85 (54.5)	'... if there is another mechanism that has led to this that we don't see here ... if maybe some marine animal has bitten him, it would be considered a trauma'.	46 (29.5)	'Well I think it's an emergency, but I don't consider it a trauma, it could be a hit to the head, then it could be a trauma'.
Management	18 (66.7)	120	6 (5.0)	'... everything that is part of ATLS, even burns, form part of trauma, no?'	94 (78.3)	'ATLS names [drowning] and classifies it underneath trauma, it is for the protocolization of the attention of the patient'.	20 (16.7)	'It's serious, it must be treated quickly because it's dangerous depending on where the injury is. Because if it is ... in a dangerous spot of the body, you have to treat it quickly'.
Psychological Trauma	13 (48.1)	94	0 (0)	Not Applicable	10 (10.6)	'... there could be trauma of another style, not only with respect to the body, but also, let's say, psychological trauma that can be caused by these kinds of ... injuries'.	84 (89.4)	'... apparently a car accident, that can be that she has not suffered a physical trauma, but psychological yes she is going to have 100%'.

*Total number of times of each theme, including all associated sub-themes, was applied during coding analysis. Application number is then presented by respondent type.
Abbreviations: Advanced Trauma Life Support (ATLS).

Table 5. Sub-themes from picture pile sorting responses.

Parent theme*	Sub-theme [†]	Interviews with sub-theme N (%)	Sub-theme applications N	Representative sub-theme quote (Participant type)
		N total = 27		
Trauma Definition	Varying Severity	13 (48.1)	26	'This would be a trauma for the person, but it's not that serious'. (Layperson)
	Accident	11 (40.7)	16	'... well, this [intoxication image] is different than most of what we have seen, because the others were accidents'. (Layperson)
	Temporality of Trauma	9 (33.3)	23	'[It's a] trauma because it's an acute occurrence, no?' (Physician)
	External Agent	4 (14.8)	13	'Some use the concept of trauma as something produced by an external agent. So, if something external has affected you in some way, it is a trauma'. (First Responder)
Mechanism	RTI	23 (85.2)	37	'... a patient who has been hit by a car as a pedestrian. Obviously, it's a trauma'. (Physician)
	Burn	23 (85.2)	36	'[A] burn by liquid or could be by electricity, continues classified as a trauma ... that has to be attended to by a doctor, and that has to be treated in a hospital centre to do the final treatment'. (First Responder)
	Fall	22 (81.5)	58	'It's an elderly person. And, and, an elderly person who has fallen down stairs. Probably, eh, a cranioccephalic trauma and a hip fracture, which is most common in this kind of patient'. (Physician)
	Drowning	20 (74.1)	35	'Patient is drowning, maybe it could produce an obstruction of the airway, an asphyxiation, but it's not a trauma'. (Physician)
	GSW / Projectile	19 (70.4)	28	'... a projectile that can impact a person, and in hitting whichever part of the body it can cause ... the injury that it is going to cause is a trauma, no?' (First Responder)
	Intoxication with Drugs	18 (66.7)	27	'Because, what I see, there is a person that has been drugged, it's a psychological trauma'. (Layperson)
	Stairs	12 (44.4)	15	'... I think that this person has slipped and has fallen down the steps and has hit his head and his body ... it could be that the ribs also have broken, no? So, for me, it is a physical trauma'. (Layperson)
	Location of Impact	11 (40.7)	16	'... depending on where the bullet lies ... if it goes into the leg, I don't know if it would be trauma, but maybe if it enters the spine or something like that'. (First Responder)
	Blow / Hit	10 (37.0)	27	'... what I am seeing that there has been a strong blow, it has produced a blow to the head, break of a leg or an arm ... it is a trauma'. (Layperson)
	Intoxication with Alcohol	8 (29.6)	12	'... we are thinking that he ingested alcohol, and that generated an internal trauma in the stomach'. (First Responder)
	Interpersonal Violence	7 (25.9)	11	'... injury due to firearm ... causative of violence in this case. There exist regions where they see this much more ... assault, that provoke of course a series of penetrating injuries'. (Physician)
Physical Injury	Suicide	6 (22.2)	11	'I see a person that has ... has wanted to end their life with tablets and alcohol. I believe that it carries a physiological trauma and a physical trauma as well'. (Layperson)
	Trauma Types	16 (59.3)	41	'... exposes her to the same kind of long bone trauma, could have intra-abdominal trauma, a trauma of the intra-abdominal organs, trauma of the intra-thoracic organs'. (Physician)
	Fracture	15 (55.6)	34	'We associate the trauma in relation to a bone that is damaged, or a bone that is compromised'. (First Responder)
Management	Resuscitation	9 (33.3)	24	'... a lot of emphasis is placed on certain topics, ah, in the patency of the airway ... with respect to the management of the trauma patient'. (First Responder)

(Continued)

Table 5. Continued.

Parent theme*	Sub-theme [†]	Interviews with sub-theme	Sub-theme applications <i>N</i>	Representative sub-theme quote (Participant type)
		<i>N</i> (%) <i>N</i> total = 27		
Psychological Trauma	Assess the Scene	7 (25.9)	20	'For us, the scene is enriched with a lot of information ... we see all these details as something possible to be able to inform ... to the medical centre ... what could have happened'. (First Responder)
	Triage	6 (22.2)	19	'... we have to see what type of hospital or clinic we have to take the patient ... that have the technology to treat this patient, or the capacity'. (First Responder)
	Need for Help	6 (22.2)	15	'It's a trauma ... it requires help, no? Immediately. I believe that, I feel it needs immediate help ... if they're drowning something has to be done' (Layperson)
	Certification programmes	3 (11.1)	10	'We based ourselves in the American school, and we follow the lines of the courses of ATLS, ACLS, PALS, PHTLS'. (First Responder)
	Emotional stress	6 (22.2)	18	'... he has a physical trauma, and an emotional one as well, because it is going to leave him with a very ugly scar. If it is a leg, he is not going to ever want to wear shorts'. (Layperson)
	Fear	5 (18.5)	28	'This is a trauma because he is suffering in this, and is traumatised with a bad psychology experience. They are always going to have a fear of the water'. (Layperson)

*Each parent theme served as a unique code as well, with applications independent of the sub-theme code applications.

[†]Only sub-themes with at least 10 applications are included. A complete list can be found as supplemental material.

Abbreviations: Road Traffic Incident (RTI), Gunshot Wound (GSW), Advanced Trauma Life Support (ATLS), Advanced Cardiac Life Support (ACLS), Pediatric Advanced Life Support (PALS), Prehospital Trauma Life Support (PHTLS).

from a height ... we catalogue them as, as a possible situation of trauma'. Even a patient who is 'on their feet and fell from their own height ... it can be treated as a trauma'. A few first responders used the term 'kinematics' as the process of assessing a scene to determine if a situation is 'a possible situation of trauma'. One explained, 'The scene is enriched with a lot of information ... it gives a bigger idea ... of what could have happened' and 'what it is that the patient could have'. Lastly, a number of interviews discussed the anatomic location of impact affecting whether or not a picture represented a trauma. One first responder specified, 'Depending on where the bullet lies ... it if goes into the leg, I don't know if it would be a trauma, but maybe if it enters the spine'.

Physical injury

All of the physicians, first responders, and community members, as well as eight of 11 trauma patients discussed physical injury in relation to trauma. Respondents were most likely to reference physical injury when discussing the pictures depicting the fall and RTI. Of these respondents who discussed physical injury, nearly all affirmed the respective image represented a trauma. One remarked, 'Obviously, it's a trauma, no? Because she could have multiple traumas, and injuries in this part of her body'. Another explained, 'The fact that the person has been impacted by a vehicle exposes her to the same kind of long bone trauma; could have intra-abdominal trauma [and] trauma of the intra-thoracic organs'.

Fewer than half of respondents referenced physical injury for the drowning or intoxication images; for those who did, most explained the mechanism 'in and of itself is not trauma'. For the drowning picture, many noted 'signs of physical harm aren't seen' and went on to say the image would represent a trauma only if there was an associated physical injury from another mechanism, such as 'if a shark grabs them' or 'the patient has fallen ... from a height'. Similarly, respondents to the intoxication image stated it did not represent trauma 'because they don't have an injury,' and that it would be considered as trauma if 'the person ... is standing and fell'.

Statements about physical injury sometimes included references to orthopaedic injury; more than half of interviews mentioned fractures or musculoskeletal trauma. One first responder group stated, 'We classify trauma as that in which a person receives a heavy blow that fractures a bone'. Medical provider references to orthopaedic injury often appeared as participants reviewed possible injuries from a given mechanism, stating one 'could consider a kind of trauma or a fracture of an extremity'. For the 'elderly person who has fallen down the stairs,' one of the physicians expected to see 'a cranioccephalic trauma and a hip fracture, which is the most common in this kind of patient'. Another respondent went as far as stating the GSW picture only represented trauma 'if the bullet hits a bone and fractures it'.

Management

Aspects of trauma management were mentioned in all seven first responder interviews, four physician interviews, and seven layperson interviews. Several references were made to various training programmes and protocols, specifically the Advanced Trauma Life Support (ATLS) programme and its focus on Airway, Breathing, and Circulation (ABCs). A few participants defined trauma based on their knowledge of these programmes, stating 'ATLS names and classifies it underneath trauma,' or 'everything that is part of ATLS ... forms part of trauma'. When discussing the management of the patients depicted in the pictures, 'we follow the lines of the courses of ATLS,' 'you have to ... do all that is the ABCs of life to stabilise the patient to be able to transport them'. Some referenced the need to act quickly with traumas, stating 'it must be treated quickly because it's dangerous depending on where the injury is'. Others mentioned the need for 'control of the head ... control of the vital signs, control of the airway, and control of the spine'.

All participants who discussed management for the fall, RTI, burn, and GSW pictures, stated the images represented trauma. While a minority of respondents stated drowning was a trauma, one

respondent explained, ‘You have to treat it similarly like if you had a trauma ... not a trauma of organs, but the form of attending to the patient’. For intoxication, one first responder remarked, ‘In no way would we bring down a trauma team ... one would not have to put on a cervical collar, unless we encounter things that we can’t see in this photograph’.

Psychological trauma

Many non-physician respondents expressed that trauma is not ‘just due to a contusion, but rather it can be psychological too’. Nine layperson and four first responder interviews discussed psychological trauma. In contrast, this theme did not appear in any physician responses. This concept was described by a layperson as ‘something that I have lived and that has made me suffer ... that has caused me harm so there is a trauma, let’s say, in me’. The two pictures most often associated with psychological trauma were GSW and intoxication. One participant noted that intoxication is ‘different than most of what we have seen, because the others were accidents ... this is a person that ... wanted to take his life ... that is a psychological trauma’. Of those who thought the GSW represented trauma, four explained it was solely because of psychological trauma rather than any physical injury. ‘To see a revolver, one becomes fearful and always is going to have that image that they assaulted you ... it becomes a nightmare’.

Participants frequently categorised pictures as traumas if they perceived that the experience would result in a long-standing fear or phobia. They made this association with the drowning picture because ‘they are always going to have fear of the water’. Similarly, in reference to the fall picture, ‘psychologically it produces fear of falling again’. For the woman depicted in the RTI image, ‘each time she goes out on the street, each time she gets in a car the psychological trauma will remain’ and she will have ‘fear of the cars, or ... fear now of travelling in front’. Multiple respondents reacted similarly to the burn image, stating the depicted person would have ‘an emotional [trauma] as well’ because of the ‘very ugly scar’ and that the person is ‘not going to want to wear shorts’.

Discussion

While the definition and mechanisms of injury are well accepted, there is an absence of literature examining the understanding of the word trauma, especially across languages and cultures, and a lack of consensus on inclusion criteria for collecting trauma data. The accuracy of trauma registry data hinges on a shared understanding of which patients to include, and our study provides valuable insight on how varied meanings of trauma could impact the implementation of trauma registries, especially in settings without well-established trauma systems.

Our data demonstrate certain mechanisms, namely falls, RTIs, burns, and GSWs, were commonly categorised as trauma. Conversely, roughly half of respondents, including all of the physicians, did not consider drowning or intoxication as trauma. This inconsistency mirrors the varied trauma registry inclusion criteria in the literature; both drowning and intoxication, more commonly referred to as poisoning, are often excluded (ACS-COT, 2018).

We further examined participant understanding of trauma through a qualitative analysis of responses to the pile sorting activity. All interviews discussed mechanism, which was likely influenced by the nature of the pile sorting activity itself and the way the interviewers referred to the pictures. Even so, a majority of participants mentioned mechanism alongside the theme trauma definition, suggesting a stronger relationship than simply study design. Many went a step further to discuss additional aspects of injury mechanism, such as height of the fall or bullet trajectory. This advanced understanding of trauma was more frequent among medical personnel than laypersons, and is in agreement with widely held beliefs that trauma registry enrolment criteria should take into consideration injury severity (ACS-COT, 2018).

An unexpected theme was psychological trauma. A majority of non-physician participants mentioned the psychological aspect of trauma. Even when a participant acknowledged the clear depiction

of physical injury in an image, many also mentioned the psychological burden of the event. Moreover, there were many instances when a respondent stated an image represented trauma solely because of its psychological impact on the victim. Prior work had identified the possibility of misinterpreting the word trauma in the Bolivian context (Boeck et al., 2016), but had not uncovered the concept of psychological trauma. These varied understandings of the word trauma underscore the fact that there are nuances to translating words, surveys, or forms from one cultural and linguistic context to another (Ferrer et al., 2006; Gjersing et al., 2010), and care should be taken to assure the intended meaning is retained, or, as Peña (2007) says, to ‘ensure linguistic equivalence’.

This raises the question of whether it would be more effective to use the word injury instead of trauma, thus changing terminology to injury care systems and injury registries. The interpretation of trauma as a psychological entity may have unintended or unforeseen effects on the establishment and development of a trauma system. Skewed understanding of what constitutes trauma could affect both care access patterns and the effective delivery of trauma care, as well as the population’s participation in prevention activities. Hospital staff must understand when to include a patient in the trauma registry. They must also understand the most immediate need of a trauma patient is the management of physical, life threatening injuries. Ultimately, it must be clear that the psychological repercussions of the event are only important if the patient survives.

Trauma as physical injury was understood by nearly all respondents. This concept was discussed not only in labelling a picture as trauma, but also when explaining why certain pictures, namely drowning and intoxication, did not represent a trauma by this narrower definition. Several respondents went so far as to define trauma as being caused by an external agent, which mirrors the definition of injury (ACS-COT, n.d.). Related to physical injury was the relationship between trauma and orthopaedic injury, with over half of the respondents linking the two. This is particularly relevant in Latin America, where the word ‘traumatología’ in Spanish refers to orthopaedic surgery, which was noted to be an area of confusion during the pilot trauma registry implementation in Santa Cruz (Boeck et al., 2016). Others approached their understanding of trauma in terms of management. The theme management was most commonly discussed amongst first responder personnel, which may have been a result of pictures depicting pre-hospital scenes as well as these participants recently receiving training in the management of trauma patients through courses like ATLS.

This study reinforces the need for consistent trauma registry inclusion criteria, coupled with training of providers in what constitutes trauma. When designing a LMIC trauma registry, it is important to maximise usability, minimise added work to an already overburdened system, and ensure accurate capture of injury data in the area being studied. While the NTDB’s inclusion criteria are based on ICD tenth revision (ICD-10) diagnostic codes, these are not widely used or well-known in LMIC settings. To facilitate participation in the registry, we propose using the broadest definition to include all patients presenting with an injury as defined by the WHO. Notably, this would include both drowning and poisoning, which were not considered to be trauma by any physician in this study. In addition to specific ICD-10 diagnostic codes, the NTDB requires that patients additionally meet one of the following criteria: hospital admission, transfer to another hospital for higher level of care, or death resulting from the injury (ACS-COT, 2018). This excludes patients with superficial or minor injuries. For the purposes of implementing a new registry and compiling as complete a dataset as possible, we recommend initial stages of LMIC trauma registries include all patients who present with injury, regardless of severity. As the registry becomes more robust, these inclusion criteria can be narrowed to focus solely on more severe injuries in line with NTDB criteria.

A shared definition of trauma is not only important for the accuracy of data collection, but for the function and sustainability of the trauma system as a whole. Health care workers must understand which patients’ care should follow the trauma care algorithm, and first responders must understand which patients need to be transported to designated trauma hospitals. Likewise, laypersons, who in our study often understood trauma to be psychological, must understand why and for what conditions they should seek care at a dedicated trauma centre. For other LMICs wishing to establish trauma systems, this study shows a need to first explore the current understanding of trauma,

followed by tailored stakeholder training and public awareness campaigns to disseminate the purpose of comprehensive trauma systems to ensure ideal function and appropriate use.

There are several limitations to this study which should be addressed. First, the interviews and focus groups were carried out by three different American authors, two of whom were non-native Spanish speakers, and none of whom were Bolivian. Although each author followed a semi-structured guide to standardise the interviews, neither the guide nor included photographs were piloted prior to use. There are also likely to be differences in responses based how follow up questions were phrased or the discussion was guided. Additionally, translation from Spanish into English may have altered the intended meaning. The pictures selected for the activity likely influenced responses as well. Five of the original six pictures depict a mechanism of injury rather than the injury itself. The burn picture was originally the only one depicting the physical injury itself, until the GSW picture was changed to also depict a physical injury. Additionally, we used the term intoxication during our study, rather than the more widely accepted WHO category of poisoning. Lastly, given our small sample size from a single Bolivian city, these results are not necessarily generalisable to the country as a whole or to other LMICs. Nevertheless, the varied interpretations of trauma identified and the saturation of associated themes across multiple key stakeholder groups suggests a need to consider all possible meanings of trauma in future efforts to expand trauma systems.

Conclusion

Increased interest in combatting the LMIC injury burden has led to efforts to develop trauma systems, including the implementation of trauma registries. While trauma registry inclusion criteria in the United States are well defined via the NTDB, there is a lack of consistency for trauma registry inclusion criteria amongst LMIC trauma registries. This study highlights the varied perceptions of trauma amongst physicians, first responders, and laypersons. While most understand the association between the word trauma and certain injuries and mechanisms, there still exist varied applications of what actually represents trauma, particularly with drowning and intoxication. Additionally, those individuals lacking formal medical education often comprehend trauma to signify a psychological response to an event more so than a physical injury, a difference which may have unintended effects on trauma system development.

These results highlight the need for widespread training and public awareness efforts for all stakeholders within a trauma system. This is particularly true regarding trauma registry inclusion criteria. The initial stages of trauma registry implementation in LMICs should rely on broad inclusion criteria: all patients with an injury as defined by the WHO. Once initial data have been collected and the registry data quality confirmed within a given hospital or system, steps can then be taken to narrow its focus. Successful implementation of these registries is the foundation for the development of trauma systems in LMICs, and have the potential to meaningfully impact the global delivery of trauma patient care.

Acknowledgements

This study would not have been possible without our numerous local Bolivian collaborators, including the following physicians from HPTM, HSJD, and HJ: Dr. Pablo Mercado, Dr. Eve Kundmuller, Dr. German Toledo, Dr. Jose Carlos Camacho, Dr. Mauricio Dorn, and Dr. Ibeth Montoya. Additionally, we would like to thank the first responders, community members, and trauma patients for their time and thoughts, and for their continued belief in the value and possibility of an integrated trauma system in Bolivia. Lastly, we would like to thank David E. Victorson, PhD and Eric J. Keller, MD for their guidance through our qualitative analysis.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Northwestern University Feinberg School of Medicine under the Post-Graduate Fellowship in Global Health grant; and the Center for Surgery and Public Health, Brigham and Women's Hospital, Harvard T.H. Chan School of Public Health, and Harvard Medical School under the Arthur Tracy Cabot Fellowship grant. No funding entity was directly involved in the conduct of this study.

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